

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 649 Physically-Based Modeling — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Baraff & Witkin — Physically Based Modeling (SIGGRAPH course notes, free)	The primary text. Differential equation basics, particle systems, constraints, rigid bodies, and collision — written with unusual clarity and still the best free introduction to the field.
Matthias Müller — Ten Minute Physics (free video and code)	Short, complete, working implementations of almost every technique in this course, with the derivations. Watch the relevant episode before each module.
Bridson — Fluid Simulation for Computer Graphics (course notes, free)	The standard reference for Modules 11 and 12. The full book is worth owning; the free course notes cover the essentials.
Erin Catto — GDC talks and Box2D (free)	How constraint solvers are built in practice, from the author of Box2D. The sequential impulse material is the clearest explanation of Module 06 that exists.
Bargteil & Shinar — An Introduction to Physics-Based Animation (SIGGRAPH course, free)	A modern survey tying the whole field together. Good second voice throughout, and strong on deformables.
Müller et al. — Position Based Dynamics papers (free)	PBD and XPBD, from their authors. The primary sources for Module 04.

Module-by-module map

Module 01 — Simulation as an Initial Value Problem

Resource	What it covers for this module
Baraff & Witkin — Differential Equation Basics (free notes)	The first set of notes in the course pack, and the source of §1's framing. Short and exceptionally clear.
Ten Minute Physics — episodes 1–3	Getting a first simulation running, with working code. Do this before Module 02.
Bargteil & Shinar — An Introduction to Physics-Based Animation	The survey's opening sections cover exactly this module's framing, with a wider view of where the field has gone.
Hairer, Lubich & Wanner — Geometric Numerical Integration (chapter 1 free)	Why symplectic integrators behave as they do. Heavy going, and the first chapter's figures explain the energy curves of §3 better than anything else.

Module 02 — Numerical Integration

Resource	What it covers for this module
Baraff & Witkin — Differential Equation Basics and Implicit Methods (free notes)	Sections B and E of the course pack. The implicit cloth derivation of §3 is in the 1998 paper linked from the same page.
Ten Minute Physics — integration and stability episodes	The energy behaviour of §1 and §2, demonstrated interactively — which is far more convincing than reading about it.
Hairer, Lubich & Wanner — Geometric Numerical Integration	The authoritative account of why symplectic methods behave as they do. Chapter 1 is free and its figures are the best illustration of §2.
Baraff & Witkin — Large Steps in Cloth Simulation (1998, free)	The paper that made implicit cloth practical. Read it before Module 10.
Erin Catto — Numerical Integration (GDC, free slides)	The game-development view: why symplectic Euler and substepping beat cleverness in practice.

Module 03 — Particle Systems and Springs

Resource	What it covers for this module
Baraff & Witkin — Particle System Dynamics (free notes)	Section C of the course pack. Force generators, the architecture of §1, and spring forces done correctly.
Provot — Deformation Constraints in a Mass-Spring Model (1995, free)	The original three-spring cloth model of §3, and the strain limiting idea that Module 10 develops.
Ten Minute Physics — cloth and softbody episodes	Working mass-spring cloth with code, and a clear demonstration of the stretching problem.
Bridson, Fedkiw & Anderson — Robust Treatment of Collisions, Contact and Friction for Cloth (free)	Where mass-spring cloth has to go to become production-usable. Read before Module 10.

Resource	What it covers for this module
Robert Bridson — curl-noise for procedural flow (free)	The turbulence technique in §5. Short paper, immediately useful.

Module 04 — Constraints

Resource	What it covers for this module
Baraff — Constrained Dynamics (free notes)	Section F of the course pack. The Lagrange multiplier formulation of §1 and the global solve of §2, derived properly.
Müller et al. — Position Based Dynamics (2007, free)	The original PBD paper. Short, and the algorithm is exactly §3.
Macklin, Müller & Chentanez — XPBD (2016, free)	The compliance formulation of §4. Read it immediately after the PBD paper.
Müller et al. — Small Steps in Physics Simulation (2019, free)	The substepping result in §5. A short paper with a large practical payoff.
Erin Catto — Soft Constraints and Sequential Impulses (GDC, free)	The same ideas from the game-engine side, and the clearest explanation of warm starting.

Module 05 — Collision Detection

Resource	What it covers for this module
Ericson — Real-Time Collision Detection	The standard reference for this entire module. The author's errata and blog are free and substantial.
Gino van den Bergen — GJK materials (free)	GJK and its variants from someone who has implemented them repeatedly. The clearest free explanation of §2.
Erin Catto — GDC talks on contact manifolds and continuous collision (free)	The manifold persistence of §4 and speculative contacts of §3, from production experience.
Bridson, Fedkiw & Anderson — Robust Treatment of Collisions, Contact and Friction for Cloth (free)	The self-collision approach of §5, including cloth thickness. The standard reference.

Module 06 — Collision Response

Resource	What it covers for this module
Erin Catto — Sequential Impulses and contact solver talks (GDC, free)	The method of §3, from its author. The clearest explanation of accumulated impulse clamping and warm starting anywhere.
Baraff — Rigid Body Simulation II: Nonpenetration Constraints (free notes)	The LCP formulation of resting contact, derived properly. Heavier going than Catto and worth it.

Resource	What it covers for this module
Mirtich — Impulse-based Dynamic Simulation (thesis, free)	The impulse formulation of §1 in full, including the friction cone treatment.
Box2D source	A production contact solver you can read in an afternoon. Compare your implementation against b2ContactSolver.

Module 07 — Rigid Body Dynamics

Resource	What it covers for this module
Baraff — Rigid Body Simulation I and II (free notes)	Sections D and G of the course pack. The state formulation of §1, the inertia tensor, and the argument for storing momentum. The primary reading.
Mirtich — Fast and Accurate Computation of Polyhedral Mass Properties (free)	The exact volume-integral method of §2 for computing inertia from a mesh. Complete code in the paper.
Dzhanibekov effect — NASA ISS footage	The physical phenomenon of §4, filmed in orbit. Watch it before implementing, then match it.
Erin Catto — Numerical Methods and rigid body GDC talks (free)	The practical side: quaternion integration, renormalisation, and what goes wrong in production.

Module 08 — Articulated Bodies

Resource	What it covers for this module
Featherstone — Rigid Body Dynamics Algorithms	The reduced-coordinate algorithm of §2, from its author. His site has free tutorial material and spatial-vector notes.
Erin Catto — joints and motors (GDC, free)	The maximal-coordinate side: joint constraints, limits, and motors as implemented in a production engine.
Bullet Physics manual and source	Both formulations in one codebase — compare btGeneric6DofConstraint against the Featherstone path.
Ten Minute Physics — chains and ragdoll episodes	The chain convergence problem of §3, demonstrated interactively.

Module 09 — Deformable Solids

Resource	What it covers for this module
Sifakis & Barbi■ — FEM Simulation of 3D Deformable Solids (SIGGRAPH course, free)	The best free introduction to this module. Notes and slides covering everything from the deformation gradient to implicit integration.
Müller et al. — Meshless Deformations Based on Shape Matching (free)	The shape-matching method of §4, from its authors.

Resource	What it covers for this module
Irving, Teran & Fedkiw — Invertible Finite Elements (2004, free)	The inversion handling of §3. Essential if you want a deformable simulation that survives contact.
Müller & Gross — Interactive Virtual Materials (free)	The corotational formulation of §3, clearly presented.
Bargteil & Shinar — Introduction to Physics-Based Animation	The deformables chapters tie §1–§4 together well.

Module 10 — Cloth and Hair

Resource	What it covers for this module
Baraff & Witkin — Large Steps in Cloth Simulation (1998, free)	The implicit cloth paper of §2, including filtered constraints. Still the standard reference.
Bridson, Fedkiw & Anderson — Robust Treatment of Collisions, Contact and Friction for Cloth Animation (2002, free)	The layered collision treatment of §4, including rigid impact zones. Essential reading.
Provot — Deformation Constraints in a Mass-Spring Model (free)	The original strain limiting of §3.
Bergou et al. — Discrete Elastic Rods (free)	The standard formulation for hair and strands in §5, handling bending and twisting correctly.
Ten Minute Physics — cloth episodes	XPBD cloth with working code — the practical alternative to implicit integration.

Module 11 — Fluids I: Grid-Based

Resource	What it covers for this module
Bridson — Fluid Simulation for Computer Graphics	The standard reference for this module and the next. The free course notes cover the MAC grid, projection, and level sets completely.
Stam — Stable Fluids (1999, free)	The splitting and semi-Lagrangian advection of §2. Short, and it changed the field.
Stam — Real-Time Fluid Dynamics for Games (free, with code)	A complete 2D solver in about a hundred lines. Implement this first.
Fedkiw, Stam & Jensen — Visual Simulation of Smoke (free)	Vorticity confinement, from the paper that introduced it to graphics.
Ten Minute Physics — Eulerian fluid episodes	A working grid solver built up step by step, with code.

Module 12 — Fluids II: Particles and Hybrids

Resource	What it covers for this module
Bridson — Fluid Simulation for Computer Graphics	The PIC/FLIP chapters are the reference for §2, and the free notes cover the transfers in detail.
Müller, Charypar & Gross — Particle-Based Fluid Simulation (SPH, free)	The SPH formulation of §1 as graphics uses it.
Macklin & Müller — Position Based Fluids (2013, free)	PBF. Short, and the algorithm follows directly from Module 04.
Jiang et al. — The Material Point Method for Simulating Continuum Materials (SIGGRAPH course, free)	The definitive free introduction to §3, with code.
Tessendorf — Simulating Ocean Water (free)	The spectral ocean model of §4. Read it before simulating an ocean.

Module 13 — Shipping a Simulator

Resource	What it covers for this module
Glenn Fiedler — Fix Your Timestep! (free)	The accumulator pattern of §1, including the spiral of death and the interpolation. Short and definitive.
Glenn Fiedler — the rest of Gaffer On Games (free)	Determinism, networked physics, and lockstep. The practical counterpart to §3.
Erin Catto — Box2D source and GDC talks	Islands, sleeping, and the engineering of a shipped solver. Read b2World::Solve for the island construction of §2.
Bullet Physics manual	Production concerns: determinism caveats, sleeping thresholds, and debug drawing interfaces.
Christer Ericson — blog archive (free)	Numerical robustness and the practicalities of shipping physics code.